

Exercise 3: SQL foundationsSQL Case statements(01) The products Table

```
(1) SELECT product_name,  
        price,  
        CASE  
            WHEN price > 1000 THEN 'Expensive'  
            WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'  
            WHEN price < 100 THEN 'Budget'  
        END AS price_category  
FROM products;
```

product_name	price	price_category
Laptop	1200.00	Expensive
Phone	800.00	Mid-range
Keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

(02) The orders Table

```
SELECT customer_name,  
        amount,  
        CASE  
            WHEN amount >= 1000 THEN 'High Value'  
            WHEN amount >= 500 AND <= 999.99 THEN 'Medium Value'  
            ELSE 'Low Value'  
        END AS order_value_category  
FROM orders;
```

customer_name	amount	order value - category
Alice	150.00	Low Value
Bob	360.00	Medium Value
Charlie	999.99	Medium Value
Diana	45.50	Low Value
Ethan	1200.00	High Value

(03) The employees Table

```

(3) SELECT emp_name,
           department,
           salary,
           CASE

```

```

    WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'
    WHEN department = 'HR' AND salary > 55000 THEN 'Experienced HR'
    ELSE 'staff'

```

```

    END AS position_level

```

```

FROM employees;

```

emp_name	department	salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	staff
Lucy	Finance	45000	staff
Tom	HR	55000	staff

(04) The students Table

(4) SELECT student_name,

score,

CASE

WHEN score $\geq$ 90 THEN 'A'

WHEN score BETWEEN 80 AND 89 THEN 'B'

WHEN score BETWEEN 70 AND 79 THEN 'C'

WHEN score BETWEEN 60 AND 69 THEN 'D'

ELSE 'F'

END AS grade

FROM students;

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ellie	68	D

(05) The deliveries Table

(5) SELECT delivery_id,

delivery_time_minutes,

CASE

WHEN delivery_time_minutes $\leq$ 30 THEN 'Fast'

WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'On time'

ELSE 'Late'

AND AS performance

FROM deliveries;

delivery_id	delivery_time_minutes	performance
1	45	On time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

(06) The tickets Table

```

(4) SELECT issue_type,
           priority,
           CASE
             WHEN priority = 3 THEN 'High'
             WHEN priority = 2 THEN 'Medium'
             WHEN priority = 1 THEN 'Low'
           END AS priority_label
FROM tickets;

```

issue_type	priority	priority_label
Long issue	1	Low
Server down	3	High
Slow system	2	Medium
Email errors	2	Medium
Password reset	1	Low

(07) The attendance Table

```
(7) SELECT student_id  
      (day_present * 100.00 / total_days) AS attendance_percentage  
      CASE  
        WHEN (day_present * 100.0 / total_days) >= 90 THEN 'Excellent'  
        WHEN (day_present * 100.0 / total_days) >= 75 THEN 'Good'  
        ELSE AS 'Needs Improvement'  
      END AS attendance_status  
FROM attendance;
```

student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Needs improvement
3	98	Excellent
4	50	Needs improvement
5	100	Excellent

(08) The product inventory Table

```
(8) SELECT product_id,  
      stock_qty,  
      CASE  
        WHEN stock_qty = 0 THEN 'Out of Stock'  
        WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'  
        ELSE 'In Stock'  
      END AS stock_status  
FROM product_inventory;
```

product_id	stock_qty	stock_status
1	5	Low Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

(09) The classes Table

(9) SELECT subject,
enrolled_students,
CASE

WHEN enrolled_students $\geq$ 25 THEN 'Large'
WHEN enrolled_students $\geq$ 10 THEN 'Medium'
ELSE 'Small'

END AS class_size_category

FROM classes

subject	enrolled_students	class_size_category
Math	20	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

(10) The payments Table

```
(10) SELECT payment_id,  
           payment_method,  
           amount,  
           CASE
```

```
           WHEN payment_method = 'Cash' AND amount >= 200
```

```
           THEN 'Eligible for Discount'
```

```
           ELSE 'Not Eligible'
```

```
           END AS discount_eligibility
```

```
FROM payments;
```

payment_id	payment_method	amount	discount_eligibility
1	Cord	50.00	Not Eligible
2	Cash	200.00	Eligible for Discount
3	Cord	150.00	Not Eligible
4	PayPal	75.00	Not Eligible
5	Cash	200.00	Eligible for Discount.